# RADIO ASTRONOMY LABORATORY

Experiment: Antenna Characterization

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# 1 Aim

To characterize the given antenna by measuring received power as a function of separation and angular position in co-polarization and cross-polarization configurations, following the manual's procedure.

# 2 Objectives

1. Determine operating frequency and return loss from VSWR measurements (Part A; pending).
2. Measure the effect of distance on received power (Part B).
3. Measure angular beam patterns in co-polarization and cross-polarization (Parts C and D).
4. Determine half-power beamwidth (HPBW) and estimate directivity when the required plane-labelled patterns are available (Part E).

# 3 Theory

An antenna converts electrical power into electromagnetic radiation and vice versa. An antenna beam pattern describes how received or radiated power varies with direction. The main lobe contains the direction of maximum response; side lobes and a back lobe occur away from the main direction.

For the distance measurement, received power is recorded at different separations $R$. In free-space far-field conditions, the received power is expected to decrease approximately with the square of distance, corresponding to a trend of about $-20$ dB per decade when plotted against $20\log_{10} R$. Reflections, alignment, antenna mismatch, and near-field effects can cause deviations.

For angular measurements, the recorded received power is plotted against angle. Co-polarization means the transmitting and receiving electric-field orientations are aligned; cross-polarization means they are perpendicular. The manual defines half-power beamwidth as the angular width between the two points where the co-polarized main-lobe power is 3 dB below its maximum. The approximate directivity is

$$D \approx \frac{4\pi}{\theta_E \theta_H},$$

where $\theta_E$ and $\theta_H$ are the E-plane and H-plane HPBWs in radians.

# 4 Apparatus

Transmitting and receiving antennas, RF signal generator, spectrum analyser, antenna support/rotation arrangement, coaxial cables and RF adapters, as applicable to the manual's setup.

# 5 Experimental procedure

## 5.1 Part A: VSWR and operating frequency (pending)

The manual requires sweeping frequency, recording $V_{\max}$ and $V_{\min}$, and calculating

$$\mathrm{VSWR} = \frac{V_{\max}}{V_{\min}}, \qquad |\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}, \qquad RL = -20\log_{10}|\Gamma| \text{ dB}.$$

These measurements were not taken in this session. The table and frequency plots will be completed after the next lab.

### 5.2 Part B: Effect of distance

The separation between the antennas was varied and received power was recorded. The readings were plotted against $R$ and $20\log_{10} R$.

### 5.3 Part C: Co-polarization angular scan

The recorded angular interval was 10°. The notebook contains received-power columns $P_H$ and $P_E$; the values are retained under these recorded labels.

### 5.4 Part D: Cross-polarization angular scan

The recorded angular interval was 15°. The received-power columns $P_H$ and $P_E$ are retained as recorded.

## 6 Observations and calculations

### 6.1 Part A: VSWR data

| Frequency (MHz) | $V_{\max}$ (V) | $V_{\min}$ (V) | VSWR / return loss |
|---|---|---|---|
| *Pending: readings to be taken in the next lab session.* | | | |

### 6.2 Part B: Received power versus distance

Table 1: Measured received power at different antenna separations

| Distance $R$ (cm) | Received power (dBm) |
|---|---|
| 72 | -37.81 |
| 122 | -43.25 |
| 172 | -46.50 |
| 222 | -48.72 |
| 272 | -48.81 |
| 322 | -50.89 |
| 372 | -56.33 |
| 422 | -51.67 |
| 472 | -59.00 |
| 522 | -53.74 |
| 574 | -58.00 |

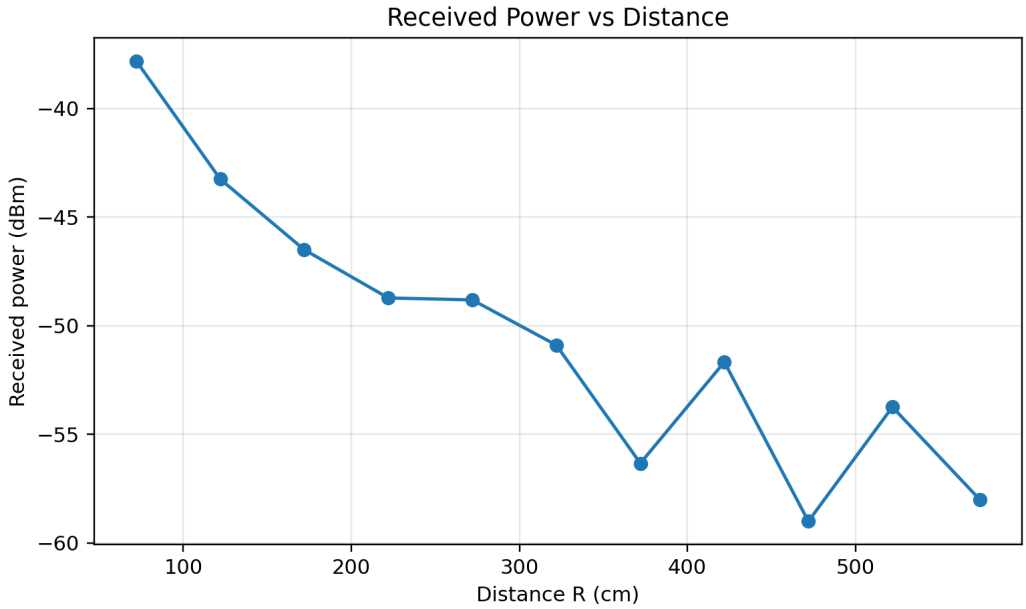



Figure 1: Received power versus separation.

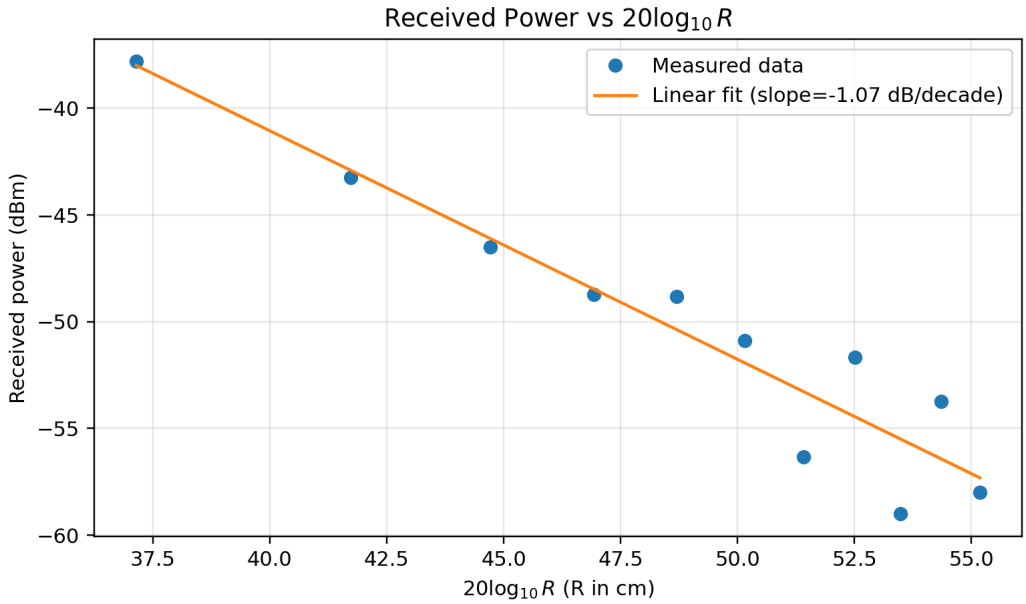



Figure 2: Received power versus $20\log_{10} R$. The fitted slope is descriptive of these measurements; deviations from the ideal free-space trend may arise from alignment and reflections.

The measured power generally decreases as separation increases, although the readings are not strictly monotonic. In particular, some points rise relative to the preceding point, which can occur because of measurement variation, reflections, or small alignment changes. The recorded data have been retained without smoothing.

### 6.3 Part C: Co-polarization angular readings

Table 2: Co-polarization readings at 10° intervals

| Angle (degrees) | $P_H$ (dBm) | $P_E$ (dBm) |
|---|---|---|
| 0 | -41.110 | -61.640 |
| 10 | -41.610 | -67.410 |
| 20 | -43.330 | -64.690 |
| 30 | -45.700 | -63.380 |
| 40 | -48.580 | -63.320 |
| 50 | -49.800 | -61.873 |
| 60 | -53.300 | -61.200 |
| 70 | -58.040 | -62.260 |
| 80 | -58.770 | -65.470 |
| 90 | -67.100 | -71.240 |
| 100 | -64.300 | -76.540 |
| 110 | -74.050 | -74.290 |
| 120 | -83.500 | -67.500 |
| 130 | -73.200 | -64.410 |
| 140 | -62.000 | -66.200 |
| 150 | -61.900 | -68.120 |
| 160 | -61.160 | -69.270 |
| 170 | -60.040 | -73.750 |
| 180 | -62.000 | -71.050 |
| 190 | -64.500 | -64.570 |
| 200 | -60.400 | -67.790 |
| 210 | -58.800 | -68.140 |
| 220 | -58.950 | -66.840 |
| 230 | -58.490 | -74.560 |
| 240 | -60.880 | -72.300 |
| 250 | -62.600 | -67.870 |
| 260 | -64.450 | -63.070 |
| 270 | -74.520 | -66.910 |
| 280 | -74.150 | -69.670 |
| 290 | -61.780 | -63.130 |
| 300 | -58.410 | -61.790 |
| 310 | -55.900 | -60.970 |
| 320 | -52.470 | -58.530 |
| 330 | -49.150 | -57.820 |
| 340 | -46.100 | -56.770 |
| 350 | -43.430 | -58.450 |
| 360 | -42.530 | -56.330 |

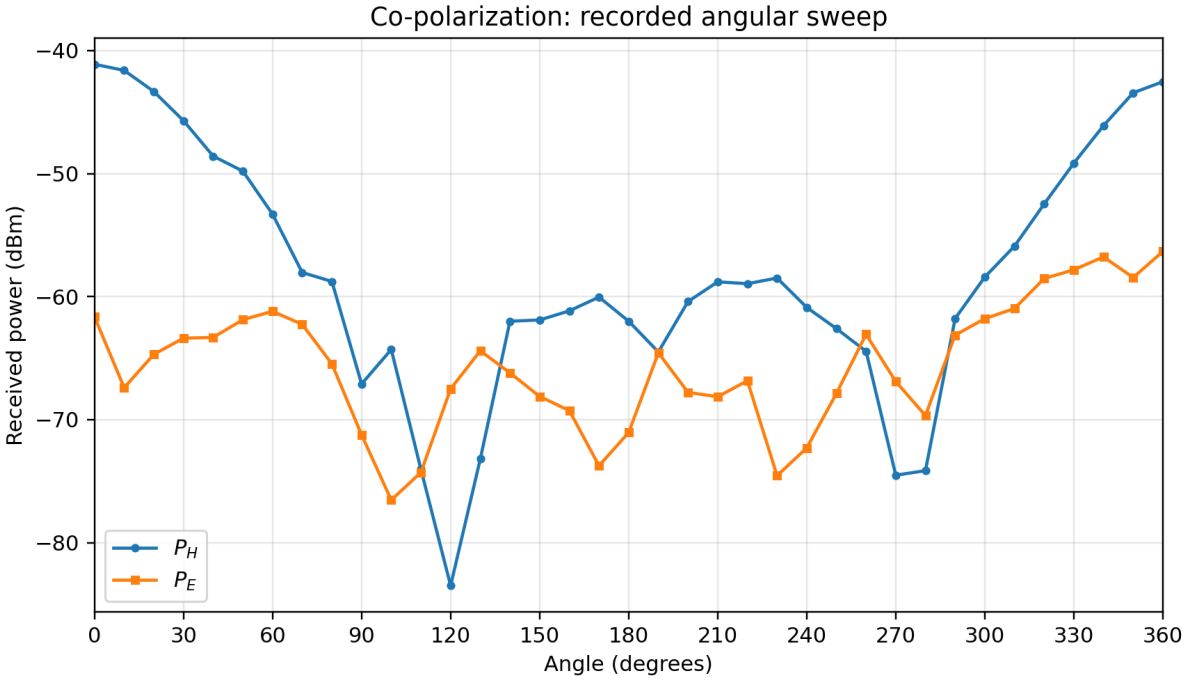



Figure 3: Recorded co-polarization angular sweep in dBm.

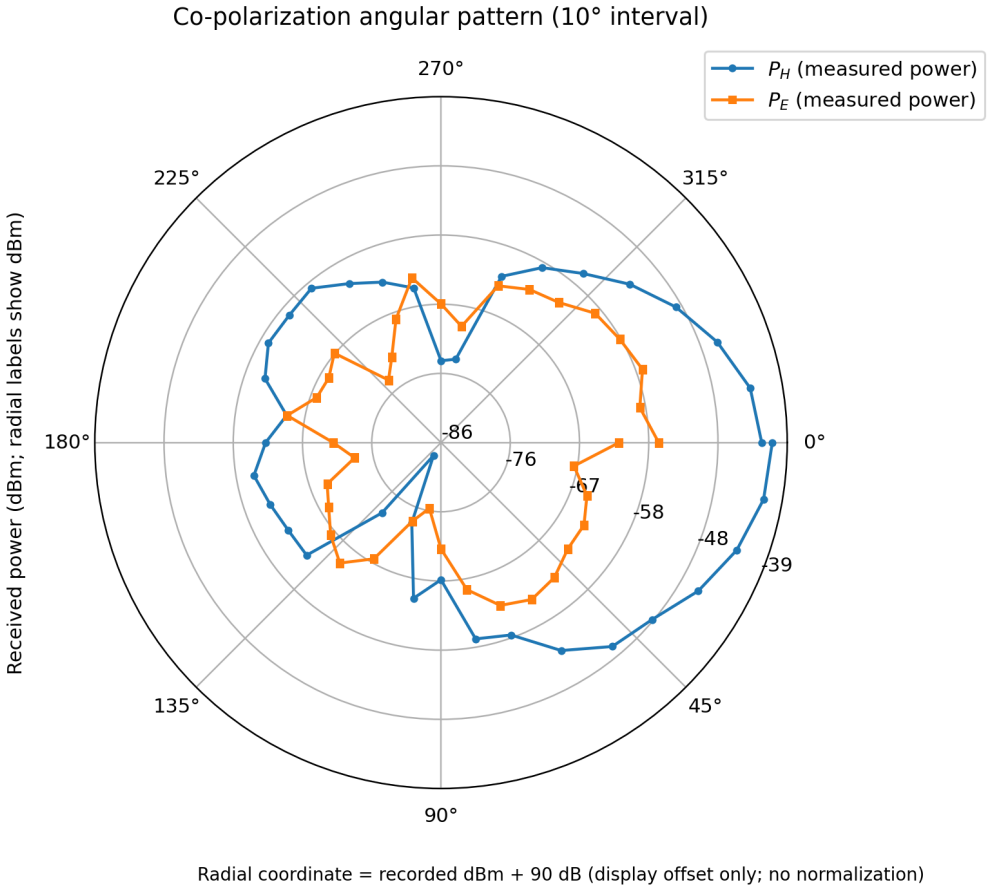



Figure 4: Co-polarization polar plot of the measured $P_H$ and $P_E$ readings; raw dBm values are shown with a constant +90 dB radial offset to make all radii positive; no normalization was applied.

### 6.4 Part D: Cross-polarization angular readings

Table 3: Cross-polarization readings at 15° intervals

| Angle (degrees) | $P_H$ (dBm) | $P_E$ (dBm) |
|---|---|---|
| 0 | -50.550 | -44.780 |
| 15 | -51.830 | -44.860 |
| 30 | -55.270 | -46.110 |
| 45 | -61.320 | -50.080 |
| 60 | -71.600 | -53.900 |
| 75 | -70.420 | -58.100 |
| 90 | -62.860 | -61.230 |
| 105 | -66.110 | -57.720 |
| 120 | -70.970 | -62.120 |
| 135 | -66.540 | -66.690 |
| 150 | -76.500 | -66.150 |
| 165 | -71.560 | -62.700 |
| 180 | -69.490 | -63.270 |
| 195 | -68.050 | -65.890 |
| 210 | -63.820 | -65.900 |
| 225 | -73.740 | -65.230 |
| 240 | -77.350 | -63.600 |
| 255 | -71.470 | -61.000 |
| 270 | -63.140 | -59.800 |
| 285 | -58.350 | -61.400 |
| 300 | -58.700 | -59.220 |
| 315 | -56.350 | -51.280 |
| 330 | -55.020 | -48.780 |
| 345 | -52.970 | -45.050 |
| 360 | -50.730 | -44.330 |

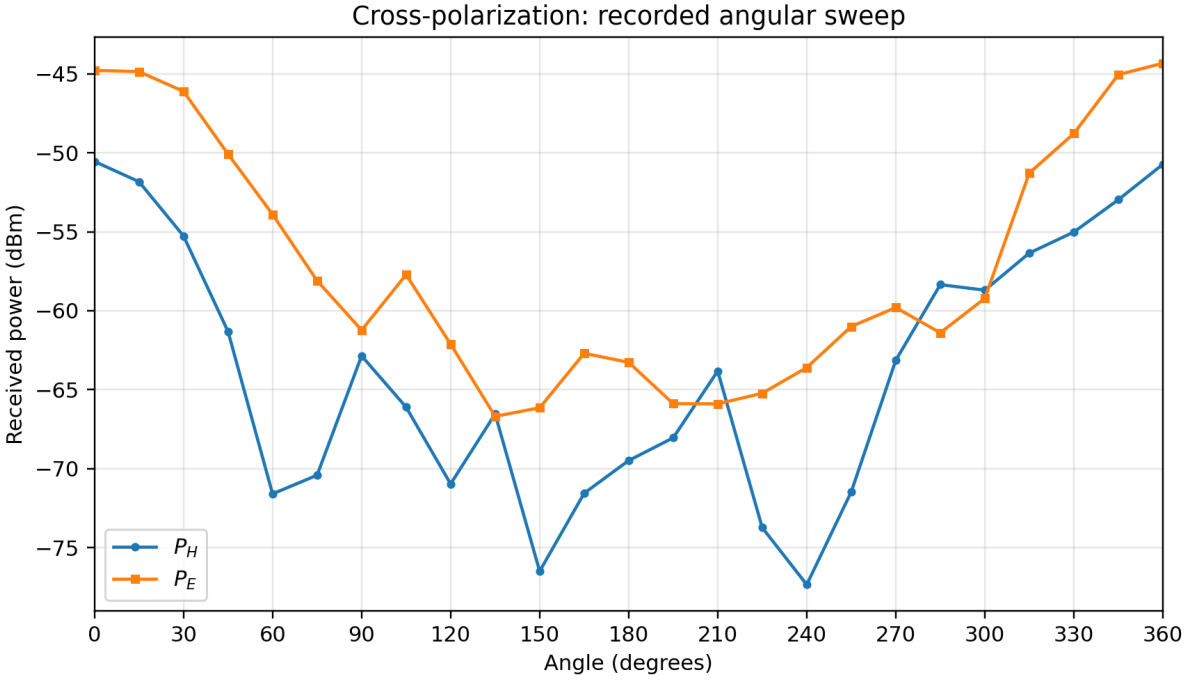



Figure 5: Recorded cross-polarization angular sweep in dBm.

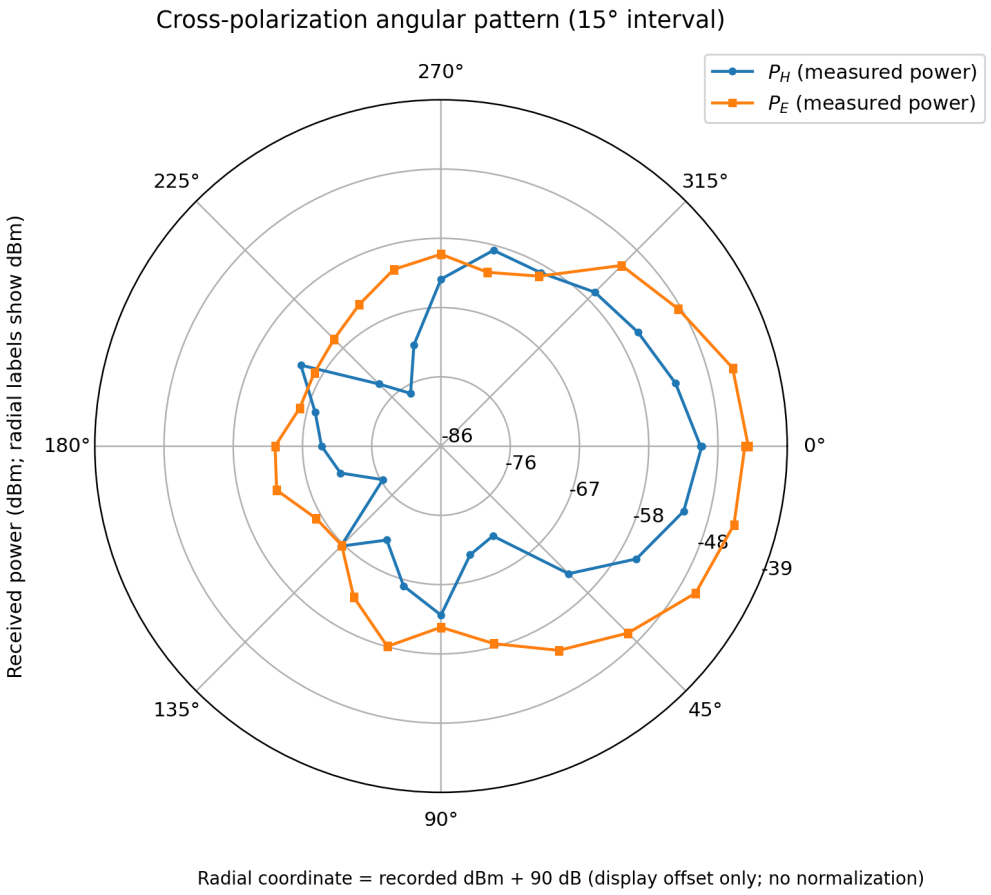



Figure 6: Cross-polarization polar plot of the measured $P_H$ and $P_E$ readings; raw dBm values are shown with a constant +90 dB radial offset to make all radii positive; no normalization was applied.

### 6.5 Part E: HPBW and directivity

The manual requires the two $-3$ dB points on each co-polarized plane pattern. The supplied notebook tables identify columns as $P_H$ and $P_E$, but do not unambiguously identify independent E-plane and H-plane co-polarized scans. Therefore, separate values of $\theta_E$, $\theta_H$, and directivity are left pending rather than assigning the polarization columns to planes without evidence. Once the E-plane and H-plane scans are confirmed, the HPBWs can be read at 3 dB below each main-lobe peak and substituted into $D \approx 4\pi/(\theta_E \theta_H)$ with angles in radians.

## 7 Results and discussion

- Part A: VSWR, return loss, operating frequency, and the range with VSWR $\leq 2$ are pending the next lab session.
- Part B: received power was measured over distances from 72 cm to 574 cm. The broad trend is decreasing received power with increasing distance, with non-monotonic fluctuations in the measured data.
- Parts C and D: angular received-power data were recorded at 10° and 15° intervals, respectively, for the columns $P_H$ and $P_E$. Both dBm-versus-angle plots and polar plots using a constant +90 dB radial offset on the recorded dBm values (without normalization) are included.
- Part E: HPBW and directivity are pending confirmation of which recorded scan corresponds to each manual-defined E-plane and H-plane pattern.

# 8 Conclusion

The distance dependence and angular response of the antenna were recorded and plotted using the supplied measurements. The measured power broadly decreases with separation, while angular scans show direction-dependent response. Since Part A was not performed and the plane assignment needed for Part E is not unambiguous in the available notebook, VSWR-based operating characteristics, HPBW, and directivity are left for completion after the missing measurements are available.